Evidence report

Environmental stressors, financial vulnerability, and social reward

Katie Social Doors | Results-ready evidence | November 7, 2026

Frozen N=97. Exploratory, not preregistered. Financial-score documentation remains provisional. This report organizes evidence for manuscript planning; it is not a draft Results section.

Scientific question	Supported finding	Interpretive boundary
1. Environment, support, and financial vulnerability	Corrected PM2.5 × MSPSS predicts FEVS and persists across ADI/ECog checks.	Conditional association; OAFEM is distinct; no mediation or causal inference.
2. Financial vulnerability and neural reward	Original FEVS–VS PPI association has opposing social/nonsocial components.	Adding corrected MSPSS yields no corrected cluster. Activation is also sensitive to adjustment.
3. Environment and neural reward	Corrected ADI × MSPSS predicts social-minus-nonsocial winning activation.	Crossover in a selected temporal region; protective buffering is not established.

What changed in the final pass? Two new whole-brain jobs add corrected MSPSS to the existing VS model. Their negative-direction map minima are .1021 (CDT2.6) and .0751 (CDT3.1), so the original corrected connectivity finding cannot be described as robust to support adjustment.

Three principal figure candidates accompany the report: behavioral conditional slopes; original VS target plus component estimates and support sensitivity; corrected ADI × support activation plus conditional slopes. Full signed/CDT inventories are in the supplementary TSV files.

Principal evidence

A compact map of findings and sensitivity

Q	Outcome / coefficient	Direction / location	Corrected evidence	Sensitivity / qualification
1	FEVS PM2.5 × corrected MSPSS	negative Behavioral; not applicable	-0.989 p=0.0005 Holm, two primary tests	HC3 b=-0.989; +ADI -0.990; +ECog -0.883; +both -0.879; all CIs exclude zero
2	Social win − nonsocial win activation FEVS at mean PM2.5	positive MNI (-31.6, 49.5, -3.7); atlas white-matter overlap	67 voxels p=0.0315 Cluster-FWE per map, CDT 3.1	+ECog p=.0505; +both .0477; standalone +ECog .0666
2	VS PPI domain-by-valence FEVS at mean PM2.5	negative Left frontal target; MNI (-20.8, 57.6, 11.2)	44 voxels p=0.0303 Cluster-FWE per map, CDT 3.1	+ADI .0320; +ECog .0349; +both .0353; +corrected MSPSS .0751 (no cluster)
3	Social win − nonsocial win activation ADI × corrected MSPSS	positive Right posterior temporal vicinity; MNI (54.8, -31.5, 8.2)	87 voxels p=0.0255 Cluster-FWE per map, CDT 3.1	Also CDT 2.6: p=.0482; LOO interaction 1.160 to 1.335; selected-region descriptive
3	VS PPI domain-by-valence ADI × ECog	positive MNI (-20.8, -72.0, -18.5)	48 voxels p=0.0473 Cluster-FWE per map, CDT 3.1	CDT 2.6 p=.0383; influence limitations; not an activation interaction
3	VS PPI domain-by-valence PM2.5 × ECog	positive / negative at separate targets Multiple targets; coordinates in supplementary cluster table	131 / 145 voxels p=.0194 / .0167 Minimum cluster-FWE per sign, CDT 3.1	Influential observations; one negative-cluster HC3 interval includes zero

Neural p values are cluster-FWE within an individual map at the stated cluster-forming threshold, not correction across exploratory models, signs, outcomes, or thresholds. Minimum p across several clusters describes that map; it does not apply to every cluster. Full precision and every cluster are retained in the supplementary tables.

Marginal relationships provide context, not model selection

Context. Describe the financial and environmental measures in the same 97-person cohort before interpreting moderation. Every support correlation below uses the corrected Island screener MSPSS mean.

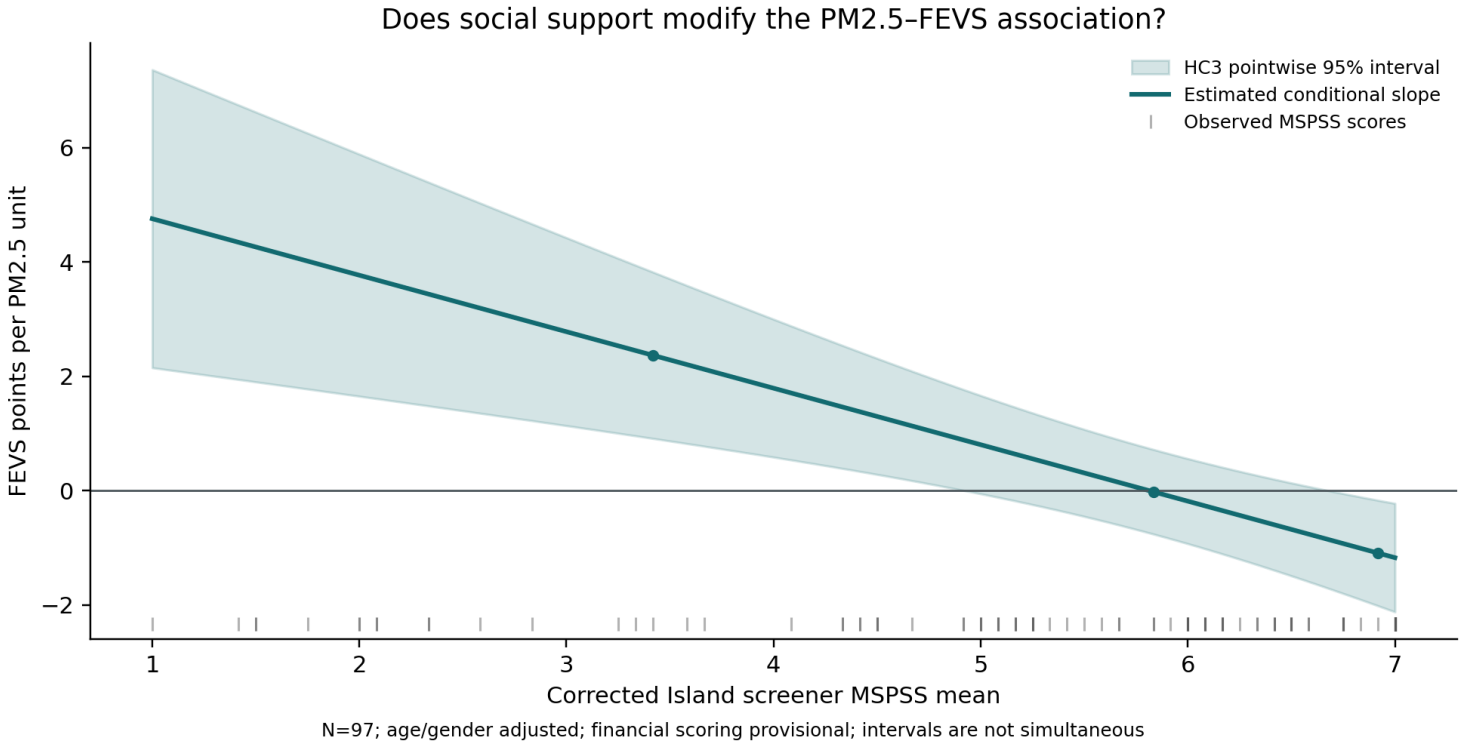
Pair	Pearson r	Spearman ρ	Pair	Pearson r	Spearman ρ
FEVS / OAFEM	0.335	0.345	OAFEM / ECog	0.458	0.326
FEVS / ADI	0.085	0.082	ADI / PM2.5	0.062	0.188
FEVS / PM2.5	0.111	0.120	ADI / MSPSS	-0.194	-0.160
FEVS / MSPSS	-0.287	-0.361	ADI / ECog	0.243	0.226
FEVS / ECog	0.393	0.297	PM2.5 / MSPSS	0.033	-0.043
OAFEM / ADI	0.162	0.121	PM2.5 / ECog	0.087	0.095
OAFEM / PM2.5	0.073	0.122	MSPSS / ECog	-0.145	-0.271
OAFEM / MSPSS	-0.121	-0.261			

N=97 for each pair. Correlations are unadjusted descriptions with no significance tests or multiplicity claims. ADI/PM2.5 are exposures, MSPSS and ECog are moderators or covariates, and FEVS and OAFEM remain distinct financial measures.

Conclusion. Weak marginal exposure–financial correlations do not exclude a conditional association. Shared variation among support, ECog, and financial scores does not identify a causal pathway.

Figure 1. The PM2.5–FEVS slope varies across observed support

Context. Does corrected perceived support modify the PM2.5–FEVS association in the existing age- and gender-adjusted model?



Content. Interaction b=-0.989; HC3 95% CI [-1.505, -0.473]; two-sided p=.000255; Holm p=.000509 across the two primary behavioral tests. The curve is a slope, not a predicted FEVS score. Bands are pointwise, not simultaneous; the rug shows observed corrected support.

Conclusion. The PM2.5 association is positive at lower support and becomes less positive, then negative, as support increases. This observational pattern is consistent with moderation; it does not demonstrate that support prevents financial vulnerability.

Observed support levels and existing sensitivity checks

Observed percentile	Corrected MSPSS	PM2.5 slope	HC3 95% CI
16	3.417	2.369	[0.913, 3.826]
50	5.833	-0.020	[-0.761, 0.720]
84	6.917	-1.091	[-2.015, -0.168]

Nearest empirical 16th/50th/84th percentiles ensure each reported moderator value is observed. Slopes use the unchanged behavioral model and its HC3 covariance, with t-based pointwise intervals (df91).

Added covariates	Interaction b	HC3 95% CI	HC3 p
baseline	-0.989	[-1.505, -0.473]	0.000255
adi	-0.990	[-1.513, -0.467]	0.000301
ecog	-0.883	[-1.406, -0.360]	0.001159
adi+ecog	-0.879	[-1.409, -0.348]	0.001435

All fits retain N=97, PM2.5, corrected MSPSS, their interaction, age, and gender. ADI and ECog enter only additively. Sensitivity p values are not corrected across these related adjustment checks.

Conclusion. The negative behavioral interaction persists after each specified adjustment. The supplied financial totals remain unchanged; stability of these fits does not validate the instrument version, scoring key, or assessment timing.

All eight behavioral tests, including secondary nulls

Outcome	Exposure × moderator	Family	Interaction b	HC3 95% CI	Holm p
FEVS	ADI × MSPSS	primary	0.010	[-0.021, 0.041]	0.522825
FEVS	ADI × ECog	secondary	0.086	[-0.031, 0.204]	0.595401
FEVS	PM2.5 × MSPSS	primary	-0.989	[-1.505, -0.473]	0.000509
FEVS	PM2.5 × ECog	secondary	1.703	[-1.777, 5.182]	0.936252
OAFEM	ADI × MSPSS	secondary	0.063	[0.004, 0.121]	0.213000
OAFEM	ADI × ECog	secondary	-0.231	[-0.682, 0.220]	0.936252
OAFEM	PM2.5 × MSPSS	secondary	-1.105	[-2.421, 0.212]	0.494552
OAFEM	PM2.5 × ECog	secondary	0.815	[-14.583, 16.212]	0.936252

Two primary tests: FEVS with ADI × MSPSS and PM2.5 × MSPSS. Six secondary tests: the two FEVS–ECog interactions and all four OAFEM interactions. Holm adjustment is separate within these original families.

Conclusion. Only PM2.5 × corrected MSPSS predicting FEVS survives its family correction. ADI × support predicting OAFEM has a nominal interaction but Holm p=.213; it is not an equivalent supported financial outcome.

Financial scoring documentation remains unresolved: administered questionnaire versions, scoring rules/keys, and source/timepoint provenance. Complete exported totals exist for all 97. No rescoring, exclusions, psychometric expansion, or cognitive-measure substitution was performed.

Task validation and neural contrasts must remain distinct

Context. The tasks should elicit ordinary reward responses before individual differences are interpreted. Existing corrected activation maps show bilateral striatal win–loss responses in both tasks.

Quantity	Definition	Role
Social winning	Social win − nonsocial win	FEVS activation and corrected ADI × support activation
Domain-by-valence	(Social win − social loss) − (nonsocial win − nonsocial loss)	FEVS VS-PPI and environmental × ECog connectivity
Task mean	Mean of social and nonsocial win−loss	Shared reward response; not social specificity

Existing anatomical bilateral VS means are 60.98 COPE units for social win–loss (descriptive 95% CI 45.02–76.94) and 72.28 for nonsocial win–loss (55.77–88.78). These anatomical summaries are descriptive checks, not new whole-brain inference.

The existing group-average domain-by-valence activation maps have no corrected clusters at either threshold or sign (best mapwise p=.2081 at CDT3.1). That null does not invalidate an individual-difference association with a different outcome or model.

Conclusion. Reward response is present in both tasks. Social-winning activation and domain-by-valence VS coupling address different questions and should not be used interchangeably.

FEVS winning activation remains secondary and qualified

Context. FEVS has a positive association with social-minus-nonsocial winning activation in the original analysis. Compare the standalone coefficient with FEVS evaluated at mean PM2.5.

FEVS coefficient	Baseline	+ADI	+ECog	+both
Standalone	p=.0411; 56 vox	p=.0458	p=.0666; no cluster	p=.0709; no cluster
At mean PM2.5	p=.0315; 67 vox	p=.0323	p=.0505; no cluster	p=.0477; 53 vox

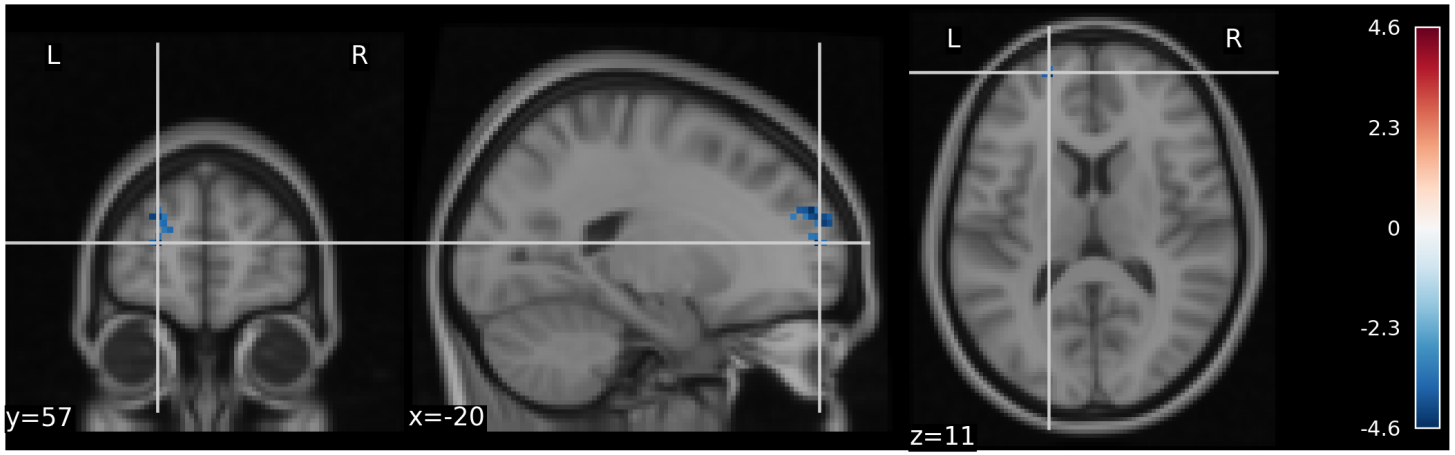
All values are positive-direction map minima at CDT3.1. Neither original coefficient passes at CDT2.6. The FEVS main coefficient at mean PM2.5 is not the FEVS × PM2.5 interaction.

Original peaks are approximately (−31.6,49.5,−6.7) standalone and (−31.6,49.5,−3.7) conditional. Harvard–Oxford subcortical atlas labels assign 79% and 66% of these clusters, respectively, to white matter; both peaks are white-matter-labeled. Atlas labels are not native tissue segmentation.

Conclusion. Describe this as a qualified left anterior/ventral frontal association, with coordinates and adjustment sensitivity. The evidence does not support a precise cortical vmPFC/OFC claim. Secondary brain figures appear in the appendix.

Figure 2A. The original VS target does not survive support adjustment

Original negative FEVS coefficient: domain-by-valence VS PPI



N=97; cluster-FWE p<.05 per map only; positive=red, negative=blue.
Same-sample selected cluster: descriptive only; no participant exclusions

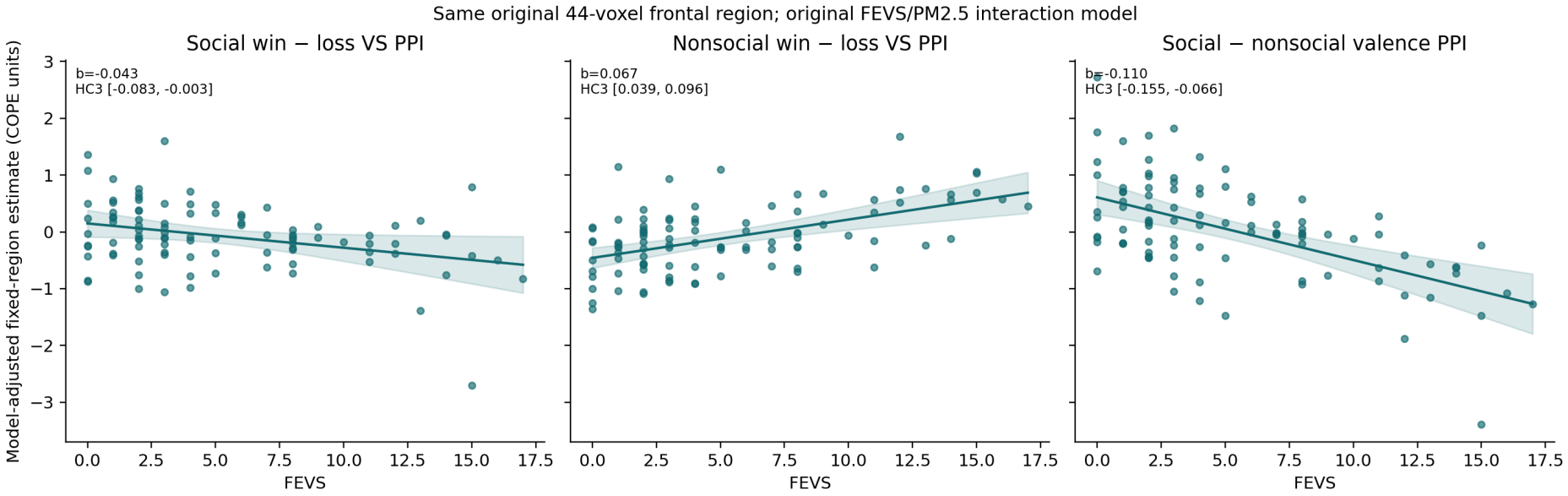
Original negative FEVS coefficient at mean PM2.5: 44-voxel left frontal target, peak (−20.8,57.6,11.2), p=.0303 at CDT3.1. The VS is the seed; the displayed frontal cluster is its task-modulated coupling target.

Covariates added to original model	Negative map, CDT2.6	Negative map, CDT3.1	Corrected clusters at CDT3.1
None	.0504	.0303	1 (44 voxels)
Corrected MSPSS only	.1021	.0751	0

Conclusion. The original whole-brain association is sensitive to perceived-support adjustment. Earlier ADI/ECog checks passed (.0320/.0349/.0353), but no cluster survives the new support sensitivity in either direction or at either threshold.

Figure 2B. Both task components contribute to the domain difference

Context. Use the original 44-voxel region as a fixed descriptive target. Fit each task component with the same original FEVS + PM2.5 + interaction and common covariates.



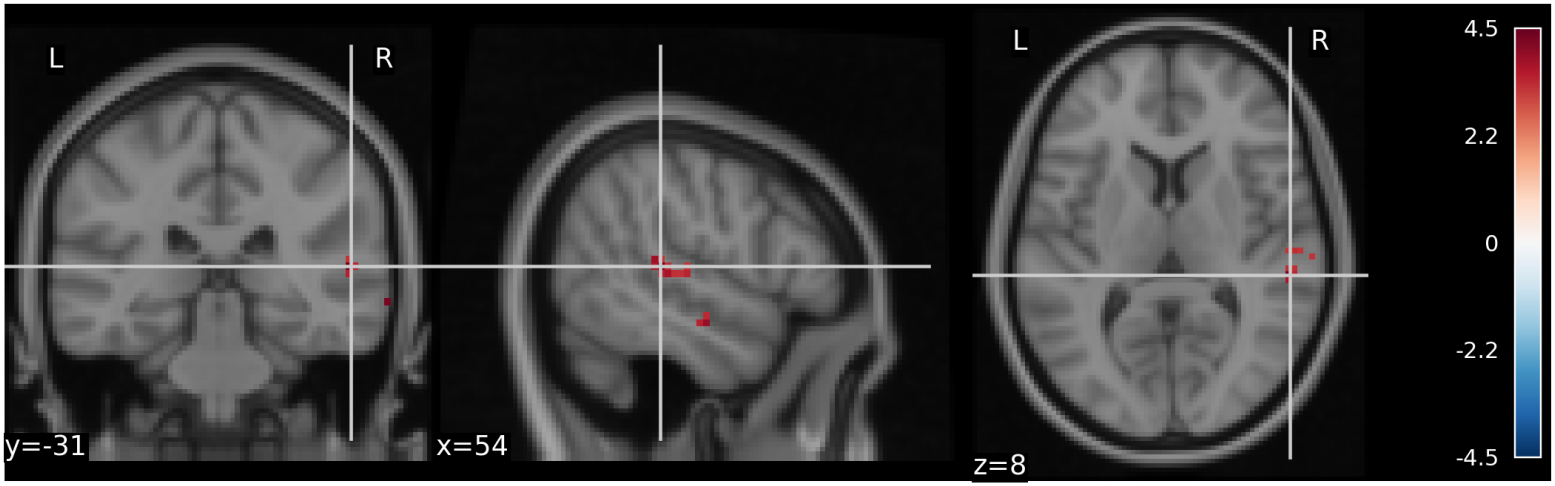
Component	FEVS slope	HC3 95% CI
Social win−loss VS PPI	-0.043	[-0.083, -0.003]
Nonsocial win−loss VS PPI	0.067	[0.039, 0.096]
Social − nonsocial difference	-0.110	[-0.155, -0.066]

Conclusion. The original negative difference reflects a negative social slope and a positive nonsocial slope. These selected-region estimates describe relative task-modulated coupling; their intervals do not independently confirm the selected whole-brain effect or establish a causal change in connectivity.

Figure 3A. Corrected ADI × support predicts winning activation

Context. Does perceived support modify the association between neighborhood disadvantage and social-minus-nonsocial winning activation?

ADI × corrected MSPSS: Social win − nonsocial win activation

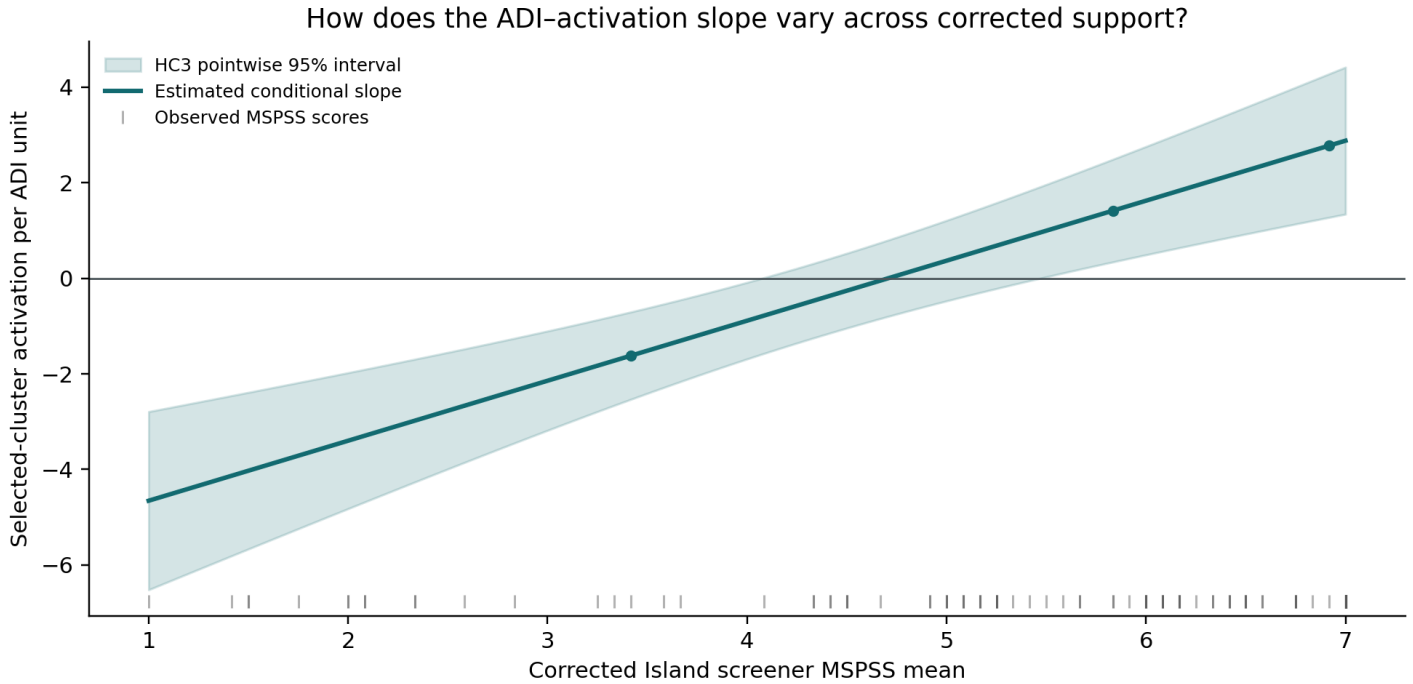


N=97; cluster-FWE p<.05 per map only; positive=red, negative=blue.
Same-sample selected cluster: descriptive only; no participant exclusions

The positive ADI × corrected MSPSS interaction has 87 voxels at CDT3.1, cluster-FWE p=.0255, peak (54.8,−31.5,8.2). The same tested interaction also passes at CDT2.6 (215 voxels, p=.0482). This is winning activation, not domain-by-valence activation or VS connectivity.

Conclusion. The cluster lies near right posterior temporal cortex. Cortical atlas overlap is 62% posterior middle temporal gyrus, 21% posterior superior temporal gyrus, and 7% planum temporale. Subcortical atlas labels include 23% white matter and a white-matter-labeled peak; anatomical precision remains limited.

Figure 3B. The selected-region pattern is a crossover



Same-sample selected cluster: descriptive only; no participant exclusions

Observed percentile	Corrected MSPSS	ADI activation slope	HC3 95% CI
16	3.417	-1.621	[-2.535, -0.706]
50	5.833	1.415	[0.340, 2.489]
84	6.917	2.775	[1.273, 4.278]

Conclusion. The selected-region ADI slope changes from negative at lower support to positive at higher support. A higher activation contrast has no established protective meaning here, so the result should not be characterized as demonstrated neural buffering.

Original activation images, model covariates, and corrected screener scores; N=97. Pointwise HC3 intervals and the rug describe a region selected in the same data. They are not independent evidence for the cluster or simultaneous moderator-wide tests.

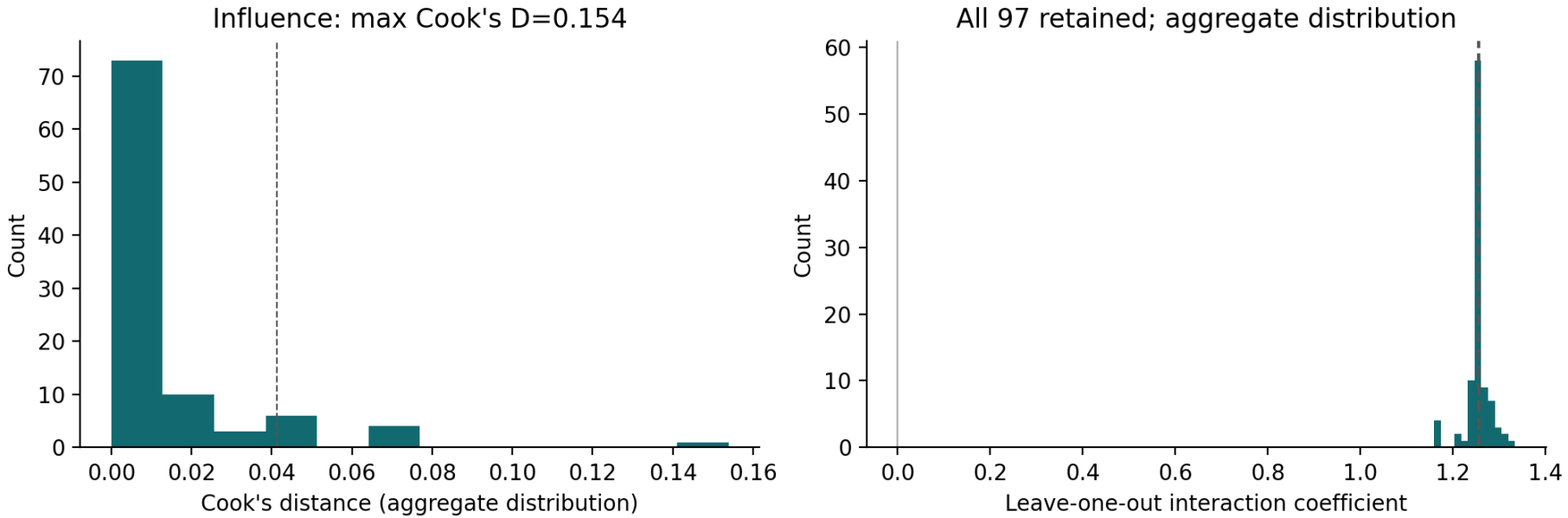
Environmental and cognitive moderation: preserve scope and nulls

Finding	Outcome / coefficient	CDT3.1 mapwise p	Interpretation
ADI × ECog	VS-PPI domain-by-valence interaction, positive	.0473 (48 voxels)	Connectivity interaction; not the corrected ADI × support activation result.
PM2.5 × ECog	VS-PPI domain-by-valence interaction, both signs	.0194 positive / .0167 negative	Multiple targets; substantial influence; one negative-cluster HC3 interval crosses zero.
ADI × ECog	VS-PPI task-mean interaction, positive	.0238 (66 voxels)	Shared-task coupling; not social specificity.
ADI × ECog	VS-PPI social−nonsocial loss interaction, negative	.0092 (3 clusters; 230 voxels)	Valence-specific follow-up; full cluster details retained.
Corrected PM2.5 × MSPSS / FEVS × MSPSS	All coefficients, outcomes, signs, thresholds	No surviving coefficients	These corrected models yield no map with a corrected cluster.

Activation ADI × ECog and PM2.5 × ECog interaction maps have no corrected clusters. ECog main effects within interaction models are conditional main coefficients; they are not evidence that the corresponding interaction is significant.

Conclusion. Retain the environmental–ECog connectivity findings as exploratory evidence with influence limitations. The corrected ADI × support winning-activation result is a separate finding; neither justifies a broad neural-buffering claim.

Selected-region influence does not require changing the cohort



Same-sample selected cluster: descriptive only; no participant exclusions

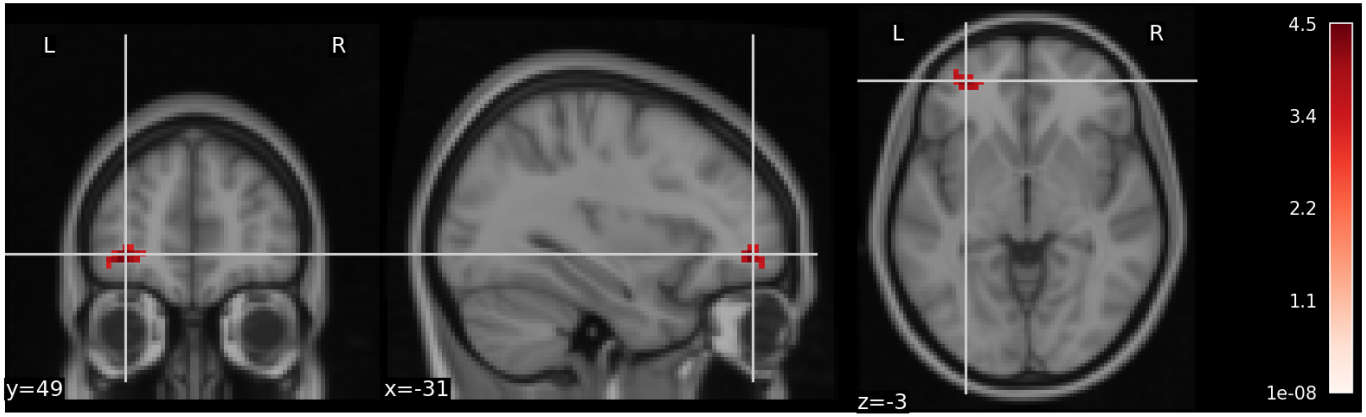
Corrected ADI × support selected-region interaction b=1.256, HC3 95% CI [0.753, 1.759]. All 97 single-case-omission coefficients remain positive, 1.160 to 1.335. Maximum Cook’s D=0.154; maximum leverage=0.252; 9 observations exceed the 4/N Cook’s-D heuristic.

The region remains fixed during these diagnostic fits. No participant is removed from the actual analysis, no exclusion is revised, and leave-one-out coefficients do not establish that a whole-brain cluster would survive rerunning permutations after omission.

Only aggregate diagnostic distributions and summaries are published. Participant identifiers, individual diagnostic rows, and source-image paths remain outside the report payload.

Secondary activation maps and influence limitations

Social win minus nonsocial win | FEVS main effect at mean PM2.5; NOT the interaction



N=97; positive FEVS coefficient; cluster-FWE p<.05; CDT Z-equivalent 3.1.
Coordinates are MNI mm. Atlas annotations describe location, not independent confirmation.

Original FEVS-at-mean-PM2.5 winning activation (67 voxels; p=.0315 at CDT3.1). The existing native map is reused; it is not recalculated by the PDF builder. Positive/negative colors follow the source legend.

Existing fixed original region	HC3 slope / interval	Single-case omission range	Max Cook's D
FEVS standalone activation	9.790 [4.036, 15.544]	7.864 to 10.440	.206
FEVS at mean PM2.5 activation	11.740 [4.309, 19.170]	9.703 to 12.661	.255
FEVS original VS difference	−.110 [−.155, −.066]	−.114 to −.098	.200

Selected-region slopes describe the original regions; they do not override whole-brain nulls after adjustment. The existing matched adjustment maps and their overlap/atlas tables remain linked in results/review/fevs-adjustment-maps-v1/.

Additional financial, ECog, and OAFEM findings remain secondary

Existing coefficient / contrast	Cluster evidence	Selected-region influence
FEVS × PM2.5; social−nonsocial loss activation	Negative; CDT3.1 p=.0495; 90 voxels; peak (−26.2,−42.3,61.7)	HC3 interaction −22.742 [−34.577,−10.906]; max Cook's D=.245; not the FEVS main coefficient during winning.
ECog at mean PM2.5; social−nonsocial loss activation	Negative; CDT2.6 p=.0318; 290 voxels	HC3 interval [−352.088,24.003] includes zero; max Cook's D=1.637. Main coefficient, not PM2.5 × ECog.
ECog at mean OAFEM; social−nonsocial winning activation	Negative; CDT3.1 p=.0453; 61 voxels	Max Cook's D=.665; one-case omissions change magnitude. Main coefficient, not OAFEM × ECog.
Corrected OAFEM × MSPSS; social−nonsocial loss activation	Positive; CDT2.6 p=.0415; 273 voxels	Corrected interaction, secondary; no new selected-region refit performed here.
Corrected OAFEM main coefficient at mean support; task-mean VS PPI	Negative; p=.0089/.0071 at CDT2.6/3.1	Distinct financial measure and common-task outcome, not FEVS social winning.
Corrected OAFEM main coefficient at mean support; social−nonsocial loss VS PPI	Positive; p=.0035/.0123 at CDT2.6/3.1	4/2 clusters respectively; full coordinates and p values retained.
Corrected OAFEM × MSPSS; task-mean VS PPI	Negative; CDT3.1 minimum p=.0349; 2 clusters, 119 voxels	Interaction distinct from the negative OAFEM main coefficient.

The full map/cluster inventory retains OAFEM × ADI/PM2.5/ECog and all other tested coefficients, including nulls. Minimum p values do not imply every cluster has the same p. No weak threshold-specific result is promoted to a principal figure.

Environmental × ECog influence summaries

Existing selected region	Coefficient	HC3 95% CI	LOO minimum / maximum	Max Cook's D
social_specific / adixecog / adixecog / z-2.6-positive-c1	0.041	[0.007, 0.075]	0.028 / 0.048	0.725
social_specific / adixecog / adixecog / z-3.1-positive-c1	0.067	[0.005, 0.129]	0.045 / 0.075	0.574
social_specific / pm25xecog / pm25xecog / z-2.6-positive-c1	1.818	[0.024, 3.612]	1.151 / 2.021	1.240
social_specific / pm25xecog / pm25xecog / z-2.6-positive-c2	1.855	[0.411, 3.299]	1.301 / 1.986	0.726
social_specific / pm25xecog / pm25xecog / z-2.6-negative-c1	-1.412	[-2.796, -0.027]	-1.769 / -0.909	1.007
social_specific / pm25xecog / pm25xecog / z-3.1-positive-c1	2.096	[0.065, 4.127]	1.387 / 2.332	0.961
social_specific / pm25xecog / pm25xecog / z-3.1-positive-c2	2.130	[0.394, 3.867]	1.399 / 2.313	0.853
social_specific / pm25xecog / pm25xecog / z-3.1-negative-c1	-1.823	[-3.596, -0.050]	-2.302 / -1.222	0.835
social_specific / pm25xecog / pm25xecog / z-3.1-negative-c2	-1.930	[-4.129, 0.268]	-2.216 / -0.953	1.875
average / adixecog / adixecog / z-2.6-positive-c1	0.025	[0.005, 0.046]	0.017 / 0.029	0.973
average / adixecog / adixecog / z-3.1-positive-c1	0.026	[0.001, 0.050]	0.016 / 0.029	1.282

These existing same-sample selected-region summaries are reused without participant exclusions. The PM2.5 × ECog clusters include Cook's D values above 1; one negative cluster's HC3 interval crosses zero. Stable single-case-omission signs are not evidence of independent validation, simultaneous-omission stability, or cluster-corrected survival after refitting.

Reproducibility, acquisition limits, and score provenance

Reused inference. Original canonical catalog: 1,328 signed/CDT maps. Exactly 384 historical support maps are replaced with 384 corrected maps; all other entries remain numerically identical. There are 71 maps with clusters, versus the historical 69. Overlapping thresholds, coefficients, and outcomes are not independent discoveries.

New inference only. Two VS-PPI domain-by-valence randomise jobs; 10,000 permutations, seed20261101, both FEVS directions; CDTs Z2.6/3.1 converted to t using df86. Original ten columns are retained exactly and centered corrected MSPSS is the sole added column (rank11, N97). A two-permutation smoke test is a workflow check, not an additional scientific analysis.

Common covariates. Neural models retain signed FD and tSNR task differences (pair means for task-mean outcomes), scanner-era indicators, age, and gender. FEVS sensitivity retains FEVS + PM2.5 + FEVS × PM2.5; no FEVS × support interaction is introduced. Behavioral models retain age and gender with their original two-primary/six-secondary correction families.

Corrected support. Ryan’s November 5 revised export; the same Island screener records are preserved, including repeat screening after scan. These are not prescan retest values or averages across administrations. Corrected scores range 1–7; 82 changed and 15 were unchanged. No independent item-level rescore or prospective temporal ordering is claimed.

Integrity and publication. Linux2 passed 24 tests and checked 9,671 source/completed-output fingerprints. This builder verifies published manifests, all 384 corrected replacements, retained historical passing maps, and component-slope reconciliation. Historical null preservation relies on the sealed Linux2 inventory audit. Participant rows are not published. Historical reports remain unchanged. Financial documentation and cognitive recovery remain unresolved/deferred.

All neural correction is mapwise cluster-FWE. Pointwise HC3 selected-region intervals are descriptive and are not extra confirmatory tests. No new whole-brain component search, exclusion change, mediation model, or manuscript Results draft was produced.

Event-file audit and boundaries of the new support sensitivity

The published read-only task audit inspected all 97 task pairs. It found zero EV-versus-own-BIDS mismatches, zero BIDS-versus-resolved-source mismatches, zero shared consumed EV files, zero identical complete task-regressor sets, zero identical canonical four-EV timing sets, and no unresolved sources or inspection errors. Native PDF design screenshots matched the inspected native files.

The tasks can look similar because their win/loss sequences and timing structure are similar. The event audit verifies recorded event provenance and consumption; it does not independently verify scanner/task acquisition identity. No native screenshot is relabeled or substituted in this report.

New sensitivity, all signs	CDT2.6 map minimum p	CDT3.1 map minimum p	Corrected clusters
Negative FEVS + corrected MSPSS	.1021	.0751	0 at both thresholds
Positive FEVS + corrected MSPSS	.8368	.6146	0 at both thresholds

There is no surviving adjusted location to compare with the original 44-voxel cluster. The published aggregate package contains corrected-map minima and thresholded figures, not the unthresholded adjusted map or an adjusted fixed-ROI coefficient. It therefore does not establish persistence of the original frontal peak, spatial pattern, or ROI slope after support adjustment. A null corrected map alone also does not establish a reversal of direction.

Published event provenance audit | All sensitivity maps

All maps with corrected clusters (1–20 of 71)

Family	Contrast	Model	Tested coefficient	Sign	CDT	Min p	Voxels	Peak MNI*
activation	Task mean	FEVS × ECog	FEVS × ECog	positive	2.6	0.0416	278	-4.6, -66.6, 14.1
activation	Task mean	OAFEM × ECog	ECog	negative	2.6	0.0250	552	33.2, -85.5, 14.1
activation	Task mean	OAFEM × ECog	ECog	negative	3.1	0.0069	161	33.2, -85.5, 14.1
activation	Winning	ADI × MSPSS	ADI × MSPSS	positive	2.6	0.0482	215	54.8, -31.5, 8.2
activation	Winning	ADI × MSPSS	ADI × MSPSS	positive	3.1	0.0255	87	54.8, -31.5, 8.2
activation	Winning	FEVS	FEVS	positive	3.1	0.0411	56	-31.6, 49.5, -6.7
activation	Winning	FEVS × PM2.5	FEVS	positive	3.1	0.0315	67	-31.6, 49.5, -3.7
activation	Winning	OAFEM × ECog	ECog	negative	2.6	0.0492	225	11.6, -45.0, 73.5
activation	Winning	OAFEM × ECog	ECog	negative	3.1	0.0453	61	-58.6, -63.9, 20.1
activation	Losing	PM2.5 × ECog	ECog	negative	2.6	0.0318	290	54.8, -15.3, 37.9
activation	Losing	FEVS × PM2.5	FEVS × PM2.5	negative	3.1	0.0495	90	-26.2, -42.3, 61.7
activation	Losing	OAFEM × MSPSS	OAFEM × MSPSS	positive	2.6	0.0415	273	17.0, -36.9, 79.5
activation	Losing	OAFEM × ECog	ECog	negative	2.6	0.0190	369	46.7, -12.6, 46.8
activation	Losing	OAFEM × ECog	ECog	negative	3.1	0.0138	116	46.7, -12.6, 46.8
connectivity	Domain × valence	ADI × ECog	ADI × ECog	positive	2.6	0.0383	188	54.8, -63.9, 29.0
connectivity	Domain × valence	ADI × ECog	ADI × ECog	positive	3.1	0.0473	48	-20.8, -72.0, -18.5
connectivity	Domain × valence	PM2.5 × ECog	PM2.5 × ECog	positive	2.6	0.0405	383	52.1, -34.2, 29.0
connectivity	Domain × valence	PM2.5 × ECog	PM2.5 × ECog	negative	2.6	0.0301	230	-50.5, -74.7, -0.7
connectivity	Domain × valence	PM2.5 × ECog	PM2.5 × ECog	positive	3.1	0.0194	131	52.1, -34.2, 29.0
connectivity	Domain × valence	PM2.5 × ECog	PM2.5 × ECog	negative	3.1	0.0167	145	52.1, -58.5, -12.6

*Peak from the cluster with the lowest corrected p in that map; voxel count is total across all its clusters. Every individual cluster, coordinate, and p is in scientific_clusters.tsv. MSPSS models use corrected scores; all other maps are retained. This table lists overlapping mapwise results, not independent discoveries.

All maps with corrected clusters (21–40 of 71)

Family	Contrast	Model	Tested coefficient	Sign	CDT	Min p	Voxels	Peak MNI*
connectivity	Domain × valence	FEVS × PM2.5	FEVS	negative	3.1	0.0303	44	-20.8, 57.6, 11.2
connectivity	Domain × valence	FEVS × ECog	FEVS × ECog	negative	2.6	0.0284	235	49.4, -66.6, -3.7
connectivity	Domain × valence	FEVS × ECog	FEVS × ECog	negative	3.1	0.0285	65	49.4, -66.6, -3.7
connectivity	Domain × valence	OAFEM × ADI	OAFEM	negative	2.6	0.0187	390	-7.3, -90.9, 17.1
connectivity	Domain × valence	OAFEM × ADI	OAFEM × ADI	negative	2.6	0.0311	366	35.9, -20.7, 61.7
connectivity	Domain × valence	OAFEM × ECog	OAFEM	negative	2.6	0.0055	727	-4.6, -53.1, 20.1
connectivity	Domain × valence	OAFEM × ECog	ECog	positive	2.6	0.0064	943	-55.9, -66.6, 17.1
connectivity	Domain × valence	OAFEM × ECog	OAFEM × ECog	negative	2.6	0.0039	1753	-28.9, -96.3, 2.3
connectivity	Domain × valence	OAFEM × ECog	ECog	positive	3.1	0.0125	72	44.0, -82.8, 26.0
connectivity	Domain × valence	OAFEM × ECog	OAFEM × ECog	negative	3.1	0.0026	707	-34.3, -69.3, -3.7
connectivity	Task mean	ADI	ADI	negative	3.1	0.0199	99	-55.9, -7.2, 5.2
connectivity	Task mean	ADI × MSPSS	ADI	negative	3.1	0.0282	47	-55.9, -7.2, 5.2
connectivity	Task mean	ADI × ECog	ADI	negative	2.6	0.0459	135	-55.9, -7.2, 5.2
connectivity	Task mean	ADI × ECog	ADI × ECog	positive	2.6	0.0225	259	14.3, -18.0, 64.6
connectivity	Task mean	ADI × ECog	ADI	negative	3.1	0.0345	43	-55.9, -7.2, 5.2
connectivity	Task mean	ADI × ECog	ADI × ECog	positive	3.1	0.0238	66	14.3, -18.0, 64.6
connectivity	Task mean	OAFEM	OAFEM	negative	2.6	0.0097	385	14.3, -72.0, -6.7
connectivity	Task mean	OAFEM	OAFEM	negative	3.1	0.0292	59	-15.4, -72.0, -0.7
connectivity	Task mean	FEVS × ADI	ADI	negative	3.1	0.0177	100	-55.9, -7.2, 5.2
connectivity	Task mean	FEVS × ECog	FEVS × ECog	negative	2.6	0.0022	965	-1.9, -85.5, 14.1

*Peak from the cluster with the lowest corrected p in that map; voxel count is total across all its clusters. Every individual cluster, coordinate, and p is in scientific_clusters.tsv. MSPSS models use corrected scores; all other maps are retained. This table lists overlapping mapwise results, not independent discoveries.

All maps with corrected clusters (41–60 of 71)

Family	Contrast	Model	Tested coefficient	Sign	CDT	Min p	Voxels	Peak MNI*
connectivity	Task mean	FEVS × ECog	FEVS × ECog	negative	3.1	0.0028	345	-1.9, -85.5, 14.1
connectivity	Task mean	OAFEM × MSPSS	OAFEM	negative	2.6	0.0089	405	14.3, -72.0, -6.7
connectivity	Task mean	OAFEM × MSPSS	OAFEM	negative	3.1	0.0071	115	14.3, -72.0, -6.7
connectivity	Task mean	OAFEM × MSPSS	OAFEM × MSPSS	negative	3.1	0.0349	119	19.7, -90.9, -18.5
connectivity	Task mean	OAFEM × ADI	OAFEM	negative	2.6	0.0214	337	14.3, -72.0, -6.7
connectivity	Task mean	OAFEM × ADI	ADI	negative	3.1	0.0450	37	-12.7, 0.9, 46.8
connectivity	Task mean	OAFEM × ADI	OAFEM × ADI	negative	3.1	0.0280	114	-64.0, 0.9, 29.0
connectivity	Task mean	OAFEM × PM2.5	OAFEM	negative	2.6	0.0227	241	14.3, -72.0, -6.7
connectivity	Task mean	OAFEM × ECog	ECog	positive	2.6	0.0278	187	-10.0, -23.4, 64.6
connectivity	Task mean	OAFEM × ECog	ECog	positive	3.1	0.0435	38	25.1, -9.9, 58.7
connectivity	Winning	ADI × ECog	ECog	positive	3.1	0.0479	50	-28.9, -90.9, 11.2
connectivity	Winning	OAFEM × ECog	OAFEM	positive	2.6	0.0310	274	44.0, -72.0, -18.5
connectivity	Winning	OAFEM × ECog	ECog	positive	2.6	0.0487	144	3.5, -77.4, 32.0
connectivity	Winning	OAFEM × ECog	OAFEM	positive	3.1	0.0264	86	44.0, -72.0, -18.5
connectivity	Winning	OAFEM × ECog	OAFEM	negative	3.1	0.0297	67	22.4, 6.3, -3.7
connectivity	Losing	ADI × ECog	ADI × ECog	negative	2.6	0.0170	742	-45.1, -69.3, -9.6
connectivity	Losing	ADI × ECog	ADI × ECog	negative	3.1	0.0092	230	-45.1, -69.3, -9.6
connectivity	Losing	OAFEM	OAFEM	positive	2.6	0.0051	1298	0.8, -85.5, -12.6
connectivity	Losing	OAFEM	OAFEM	positive	3.1	0.0134	305	-55.9, -63.9, -6.7
connectivity	Losing	OAFEM × MSPSS	OAFEM	positive	2.6	0.0035	1356	0.8, -85.5, -12.6

*Peak from the cluster with the lowest corrected p in that map; voxel count is total across all its clusters. Every individual cluster, coordinate, and p is in scientific_clusters.tsv. MSPSS models use corrected scores; all other maps are retained. This table lists overlapping mapwise results, not independent discoveries.

All maps with corrected clusters (61–71 of 71)

Family	Contrast	Model	Tested coefficient	Sign	CDT	Min p	Voxels	Peak MNI*
connectivity	Losing	OAFEM × MSPSS	OAFEM	positive	3.1	0.0123	218	-55.9, -63.9, -6.7
connectivity	Losing	OAFEM × ADI	OAFEM	positive	2.6	0.0007	2535	-1.9, -85.5, 20.1
connectivity	Losing	OAFEM × ADI	OAFEM × ADI	negative	2.6	0.0415	372	62.9, -42.3, 49.8
connectivity	Losing	OAFEM × ADI	OAFEM	positive	3.1	0.0140	735	-55.9, -66.6, -6.7
connectivity	Losing	OAFEM × PM2.5	OAFEM	positive	3.1	0.0431	55	-55.9, -63.9, -6.7
connectivity	Losing	OAFEM × ECog	ECog	negative	2.6	0.0250	174	-45.1, 11.7, 37.9
connectivity	Losing	OAFEM × ECog	OAFEM × ECog	positive	2.6	0.0298	260	-15.4, -93.6, 20.1
connectivity	Losing	OAFEM × ECog	ECog	negative	3.1	0.0170	54	-45.1, 11.7, 37.9
connectivity	Losing	OAFEM × ECog	OAFEM × ECog	positive	3.1	0.0182	102	-15.4, -93.6, 20.1
connectivity	Winning	PM2.5 + ECog	ECog	positive	2.6	0.0497	165	3.5, -74.7, 32.0
connectivity	Domain × valence	PM2.5 + ECog	ECog	positive	3.1	0.0333	55	-37.0, -66.6, 29.0

*Peak from the cluster with the lowest corrected p in that map; voxel count is total across all its clusters. Every individual cluster, coordinate, and p is in scientific_clusters.tsv. MSPSS models use corrected scores; all other maps are retained. This table lists overlapping mapwise results, not independent discoveries.

Complete model/null coverage (1–20 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
activation	ADI	Task mean	4	0	0.1889
activation	ADI	Losing	4	0	0.5349
activation	ADI	Domain × valence	4	0	0.1114
activation	ADI	Winning	4	0	0.3780
activation	ADI × ECog	Task mean	12	0	0.1162
activation	ADI × ECog	Losing	12	0	0.3007
activation	ADI × ECog	Domain × valence	12	0	0.1305
activation	ADI × ECog	Winning	12	0	0.2027
activation	ADI × MSPSS	Task mean	12	0	0.2052
activation	ADI × MSPSS	Losing	12	0	0.2483
activation	ADI × MSPSS	Domain × valence	12	0	0.0870
activation	ADI × MSPSS	Winning	12	2	0.0255
activation	FEVS	Task mean	4	0	0.4501
activation	FEVS	Losing	4	0	0.4338
activation	FEVS	Domain × valence	4	0	0.4607
activation	FEVS	Winning	4	1	0.0411
activation	FEVS × ADI	Task mean	12	0	0.2028
activation	FEVS × ADI	Losing	12	0	0.1727
activation	FEVS × ADI	Domain × valence	12	0	0.1257
activation	FEVS × ADI	Winning	12	0	0.0672

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv

Complete model/null coverage (21–40 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
activation	FEVS × ECog	Task mean	12	1	0.0416
activation	FEVS × ECog	Losing	12	0	0.2338
activation	FEVS × ECog	Domain × valence	12	0	0.2135
activation	FEVS × ECog	Winning	12	0	0.0820
activation	FEVS × MSPSS	Task mean	12	0	0.2704
activation	FEVS × MSPSS	Losing	12	0	0.2613
activation	FEVS × MSPSS	Domain × valence	12	0	0.2138
activation	FEVS × MSPSS	Winning	12	0	0.1201
activation	FEVS × PM2.5	Task mean	12	0	0.4424
activation	FEVS × PM2.5	Losing	12	1	0.0495
activation	FEVS × PM2.5	Domain × valence	12	0	0.1735
activation	FEVS × PM2.5	Winning	12	1	0.0315
activation	OAFEM	Task mean	4	0	0.5891
activation	OAFEM	Losing	4	0	0.4646
activation	OAFEM	Domain × valence	4	0	0.7963
activation	OAFEM	Winning	4	0	0.5715
activation	OAFEM × ADI	Task mean	12	0	0.1494
activation	OAFEM × ADI	Losing	12	0	0.3546
activation	OAFEM × ADI	Domain × valence	12	0	0.1166
activation	OAFEM × ADI	Winning	12	0	0.1594

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv

Complete model/null coverage (41–60 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
activation	OAFEM × ECog	Task mean	12	2	0.0069
activation	OAFEM × ECog	Losing	12	2	0.0138
activation	OAFEM × ECog	Domain × valence	12	0	0.2414
activation	OAFEM × ECog	Winning	12	2	0.0453
activation	OAFEM × MSPSS	Task mean	12	0	0.1552
activation	OAFEM × MSPSS	Losing	12	1	0.0415
activation	OAFEM × MSPSS	Domain × valence	12	0	0.1256
activation	OAFEM × MSPSS	Winning	12	0	0.0727
activation	OAFEM × PM2.5	Task mean	12	0	0.1196
activation	OAFEM × PM2.5	Losing	12	0	0.3595
activation	OAFEM × PM2.5	Domain × valence	12	0	0.1189
activation	OAFEM × PM2.5	Winning	12	0	0.2355
activation	PM2.5	Task mean	4	0	0.5878
activation	PM2.5	Losing	4	0	0.6224
activation	PM2.5	Domain × valence	4	0	0.1319
activation	PM2.5	Winning	4	0	0.3932
activation	PM2.5 + ECog	Losing	8	0	0.2038
activation	PM2.5 + ECog	Domain × valence	8	0	0.1640
activation	PM2.5 + ECog	Winning	8	0	0.1642
activation	PM2.5 × ECog	Task mean	12	0	0.0660

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv

Complete model/null coverage (61–80 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
activation	PM2.5 × ECog	Losing	12	1	0.0318
activation	PM2.5 × ECog	Domain × valence	12	0	0.1747
activation	PM2.5 × ECog	Winning	12	0	0.0791
activation	PM2.5 × MSPSS	Task mean	12	0	0.4064
activation	PM2.5 × MSPSS	Losing	12	0	0.1412
activation	PM2.5 × MSPSS	Domain × valence	12	0	0.1650
activation	PM2.5 × MSPSS	Winning	12	0	0.2747
connectivity	ADI	Task mean	4	1	0.0199
connectivity	ADI	Losing	4	0	0.1371
connectivity	ADI	Domain × valence	4	0	0.2647
connectivity	ADI	Winning	4	0	0.2256
connectivity	ADI × ECog	Task mean	12	4	0.0225
connectivity	ADI × ECog	Losing	12	2	0.0092
connectivity	ADI × ECog	Domain × valence	12	2	0.0383
connectivity	ADI × ECog	Winning	12	1	0.0479
connectivity	ADI × MSPSS	Task mean	12	1	0.0282
connectivity	ADI × MSPSS	Losing	12	0	0.0910
connectivity	ADI × MSPSS	Domain × valence	12	0	0.1244
connectivity	ADI × MSPSS	Winning	12	0	0.0873
connectivity	FEVS	Task mean	4	0	0.2177

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv

Complete model/null coverage (81–100 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
connectivity	FEVS	Losing	4	0	0.2967
connectivity	FEVS	Domain × valence	4	0	0.0805
connectivity	FEVS	Winning	4	0	0.2061
connectivity	FEVS × ADI	Task mean	12	1	0.0177
connectivity	FEVS × ADI	Losing	12	0	0.1502
connectivity	FEVS × ADI	Domain × valence	12	0	0.0997
connectivity	FEVS × ADI	Winning	12	0	0.2172
connectivity	FEVS × ECog	Task mean	12	2	0.0022
connectivity	FEVS × ECog	Losing	12	0	0.1291
connectivity	FEVS × ECog	Domain × valence	12	2	0.0284
connectivity	FEVS × ECog	Winning	12	0	0.0976
connectivity	FEVS × MSPSS	Task mean	12	0	0.1852
connectivity	FEVS × MSPSS	Losing	12	0	0.0910
connectivity	FEVS × MSPSS	Domain × valence	12	0	0.1334
connectivity	FEVS × MSPSS	Winning	12	0	0.1720
connectivity	FEVS × PM2.5	Task mean	12	0	0.1851
connectivity	FEVS × PM2.5	Losing	12	0	0.1766
connectivity	FEVS × PM2.5	Domain × valence	12	1	0.0303
connectivity	FEVS × PM2.5	Winning	12	0	0.1296
connectivity	OAFEM	Task mean	4	2	0.0097

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv

Complete model/null coverage (101–120 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
connectivity	OAFEM	Losing	4	2	0.0051
connectivity	OAFEM	Domain × valence	4	0	0.0527
connectivity	OAFEM	Winning	4	0	0.0658
connectivity	OAFEM × ADI	Task mean	12	3	0.0214
connectivity	OAFEM × ADI	Losing	12	3	0.0007
connectivity	OAFEM × ADI	Domain × valence	12	2	0.0187
connectivity	OAFEM × ADI	Winning	12	0	0.0648
connectivity	OAFEM × ECog	Task mean	12	2	0.0278
connectivity	OAFEM × ECog	Losing	12	4	0.0170
connectivity	OAFEM × ECog	Domain × valence	12	5	0.0026
connectivity	OAFEM × ECog	Winning	12	4	0.0264
connectivity	OAFEM × MSPSS	Task mean	12	3	0.0071
connectivity	OAFEM × MSPSS	Losing	12	2	0.0035
connectivity	OAFEM × MSPSS	Domain × valence	12	0	0.0512
connectivity	OAFEM × MSPSS	Winning	12	0	0.0623
connectivity	OAFEM × PM2.5	Task mean	12	1	0.0227
connectivity	OAFEM × PM2.5	Losing	12	1	0.0431
connectivity	OAFEM × PM2.5	Domain × valence	12	0	0.0987
connectivity	OAFEM × PM2.5	Winning	12	0	0.1883
connectivity	PM2.5	Task mean	4	0	0.2197

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv

Complete model/null coverage (121–134 of 134)

Family	Model	Outcome	Signed/CDT maps	Maps with clusters	Minimum map p
connectivity	PM2.5	Losing	4	0	0.4918
connectivity	PM2.5	Domain × valence	4	0	0.6963
connectivity	PM2.5	Winning	4	0	0.5745
connectivity	PM2.5 + ECog	Losing	8	0	0.1890
connectivity	PM2.5 + ECog	Domain × valence	8	1	0.0333
connectivity	PM2.5 + ECog	Winning	8	1	0.0497
connectivity	PM2.5 × ECog	Task mean	12	0	0.1789
connectivity	PM2.5 × ECog	Losing	12	0	0.1039
connectivity	PM2.5 × ECog	Domain × valence	12	4	0.0167
connectivity	PM2.5 × ECog	Winning	12	0	0.0766
connectivity	PM2.5 × MSPSS	Task mean	12	0	0.1875
connectivity	PM2.5 × MSPSS	Losing	12	0	0.1611
connectivity	PM2.5 × MSPSS	Domain × valence	12	0	0.1289
connectivity	PM2.5 × MSPSS	Winning	12	0	0.4508

Each row retains an original family/model/outcome combination, including all-null models. Map counts include tested main coefficients and interactions, both signs, and both thresholds. The minimum is descriptive, not a new omnibus test. The complete 1,328-row table gives every tested coefficient and map, including null p values: scientific_inventory.tsv